



DAT490 - Data
Science Capstone
Progress Report #2

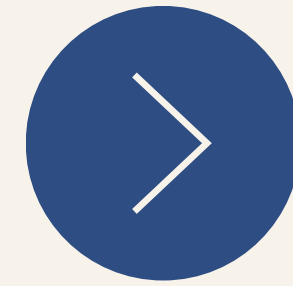


Analysis of Airline Cost Efficiencies

Jood Alaswad, Arnol Garcia, Marissa Miller, Brandon Thacker



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April 26th,
2026

Section

01

Introduction

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Introduction

The U.S. airline industry faces constant pressure to reduce costs while maintaining reliable service. Low-cost carriers are often seen as more efficient, but legacy carriers offer broader networks. We examine whether lower-cost strategies actually produce better operational outcomes — or whether they come with trade-offs in delays, cancellations, and reliability.

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Main Question:

What is the relationship between airline cost efficiency and operational performance, and are there trade-offs between cost minimization and service quality?





Research Questions

1. How do different cost structures relate to operational efficiency measures such as passengers per route and route utilization?
2. Is there a relationship between operating costs and service performance metrics such as delays, reliability, and cancellations?
3. How do low-cost, legacy, regional, and cargo carriers differ in their balance between cost, efficiency, and performance?
4. How did these relationships change across pre-COVID, COVID, and post-COVID periods?



Section

02

Data + EDA

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Major Datasets

BTS Form 41

Revenue, expenses, net
income
1,825 matched financial
observations

BTS T- 100

Passengers, freight,
route structure
4,440 carrier-quarter
observations

BTS On- Time Performance

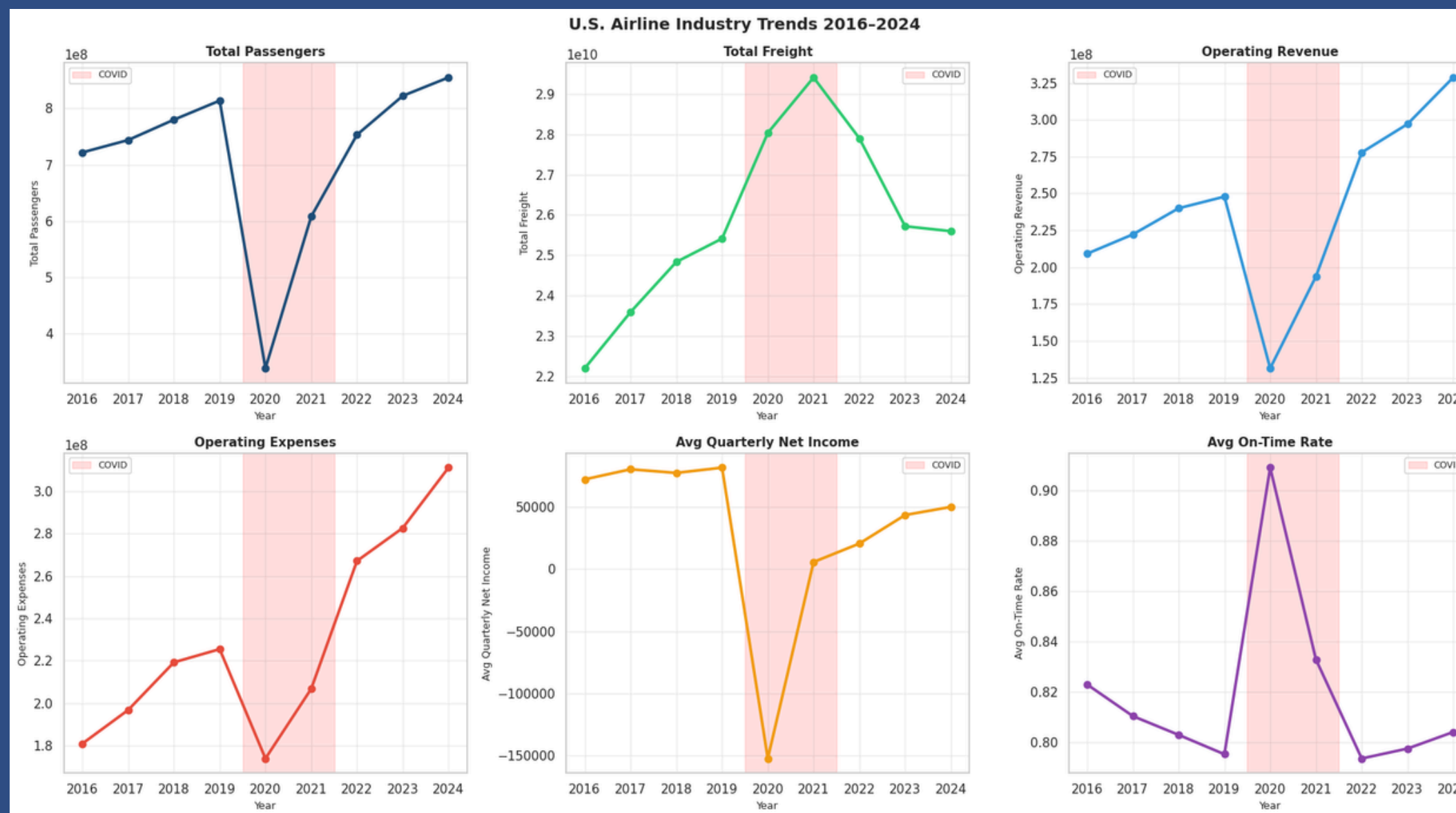
Delays, cancellations,
reliability
739 matched service
observations

681 carrier-quarter observations matched across all three datasets





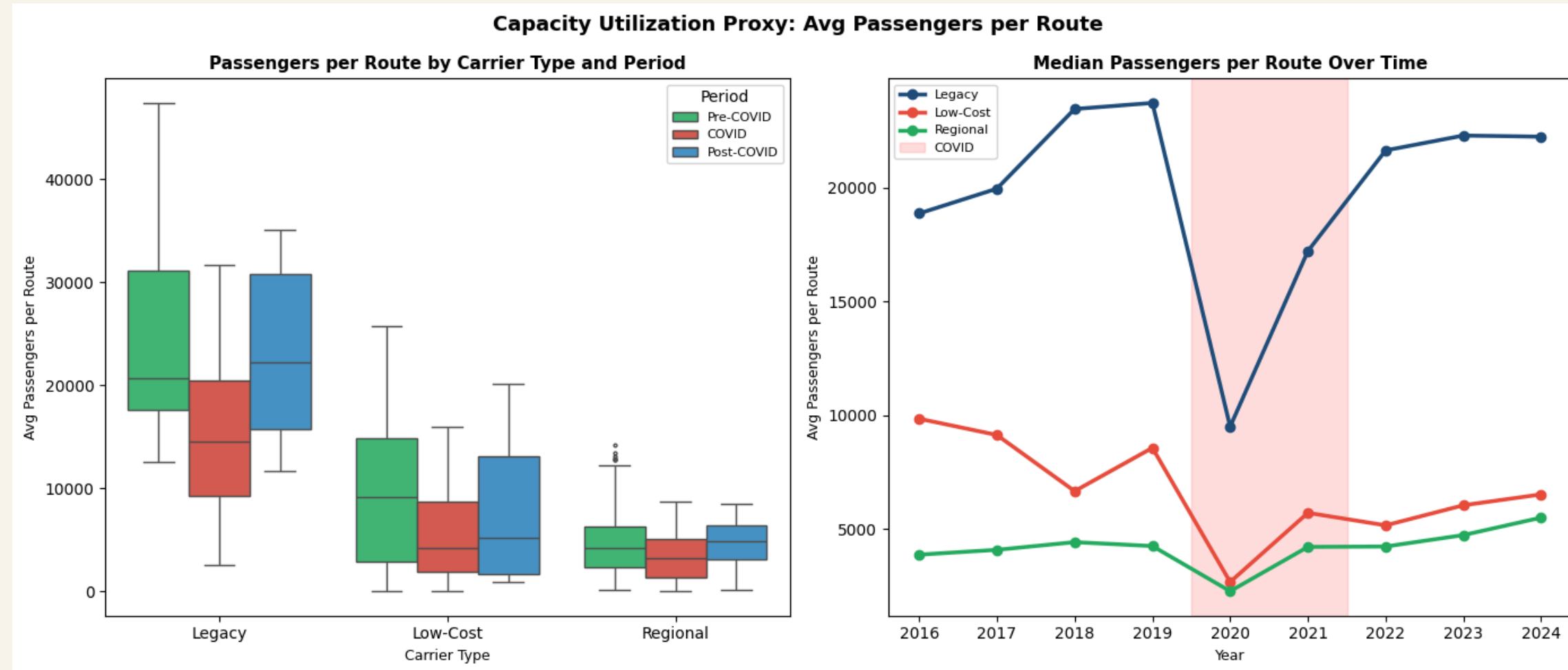
EDA: COVID



- Passenger volume peaked in 2019, collapsed in 2020, and has only partially recovered
- Freight bucked the trend – it actually increased through 2020–2021 as airlines pivoted to cargo
- Operating expenses fell more slowly than revenue, pushing average quarterly net income deeply negative (~\$150K)
- On-time rate spiked during COVID – a less congested network, not improved operations

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EDA: Carrier Type Differences



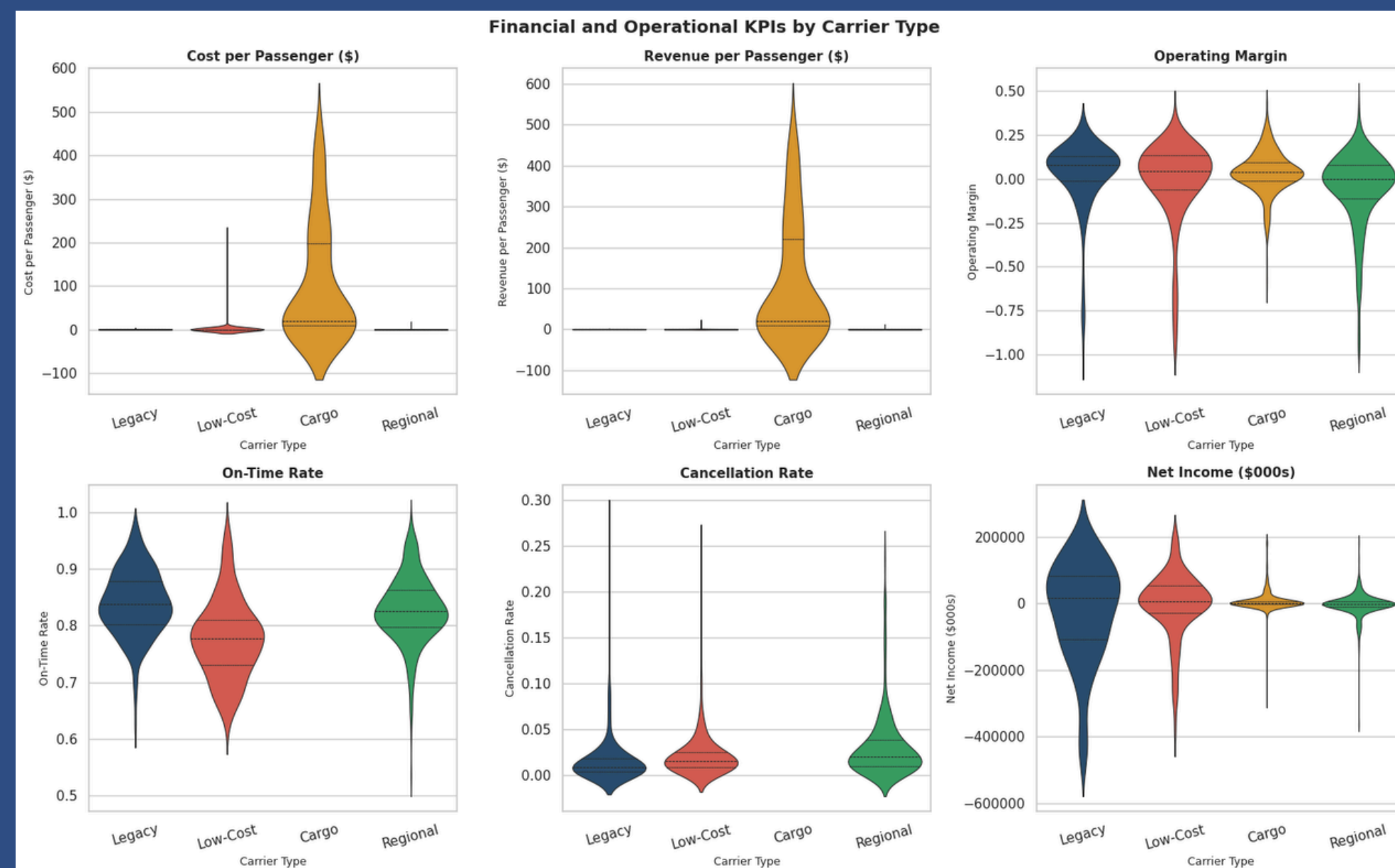
- Legacy carriers show much higher passengers per route than low-cost and regional carriers
- All groups declined during COVID, but recovery patterns differed
- This suggests network structure and traffic density matter alongside cost





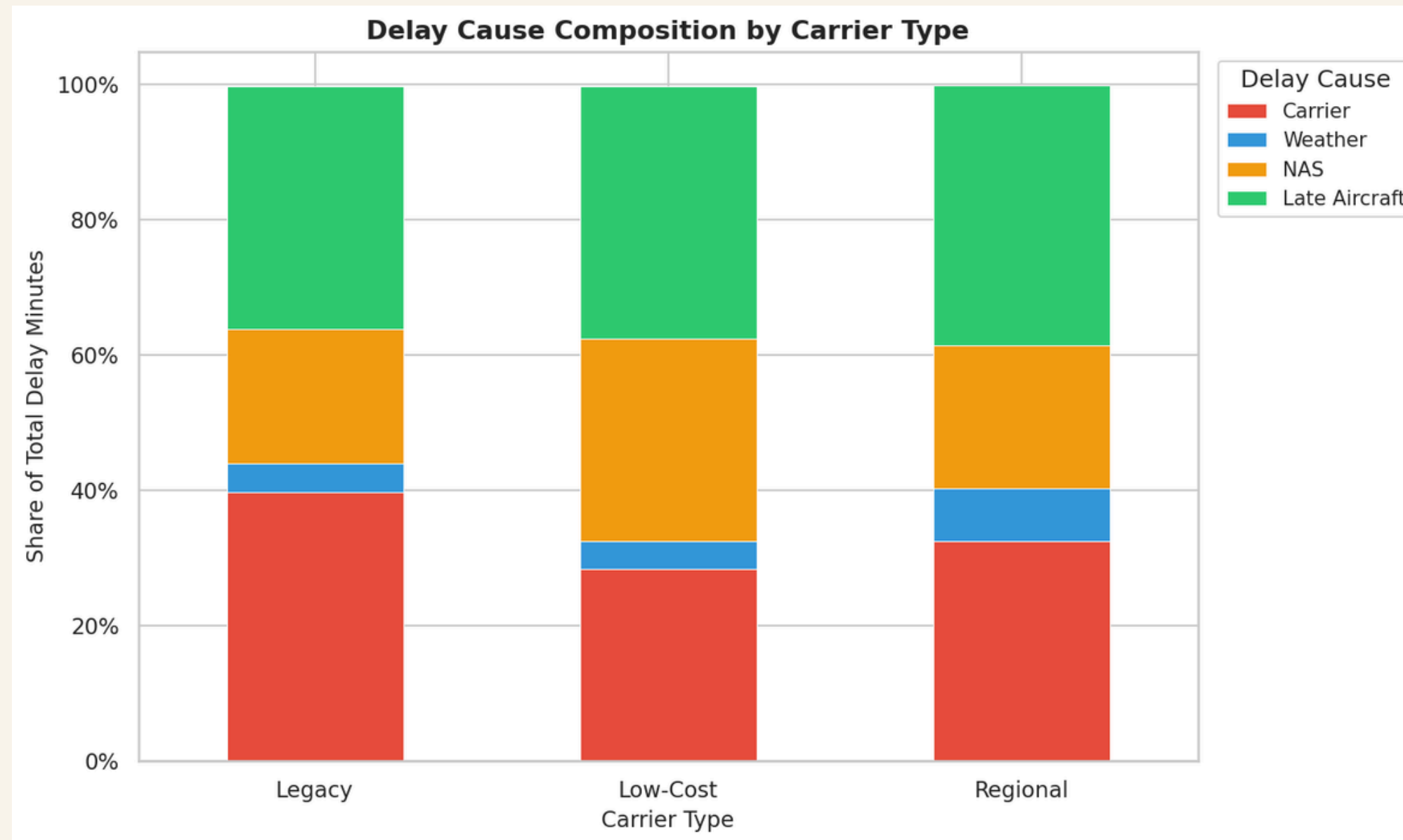
EDA: Cost and Service

- Cargo carriers show extreme cost/revenue per passenger — an artifact of near-zero passenger denominators, not a real cost signal
- Legacy carriers have wider distributional spread in margins and on-time rate; Low-Cost carriers are more tightly clustered
- Low-Cost and Legacy carriers have similar median on-time rates (~0.80), but different cancellation distributions
- **Lower cost per passenger does not reliably predict better service — business model, not cost level, is the differentiator**



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EDA: Delay Cause Breakdown



- Late-aircraft delay is the #1 source of delay minutes across all carrier types (~40%+)
- NAS delay is a larger share for Legacy and Regional — reflecting hub congestion exposure
- Carrier delay is proportionally larger for Low-Cost carriers — pointing to internal turnaround/crew factors
- **A lower on-time rate means something different depending on carrier type**



Section

03

Methodology + Early Results

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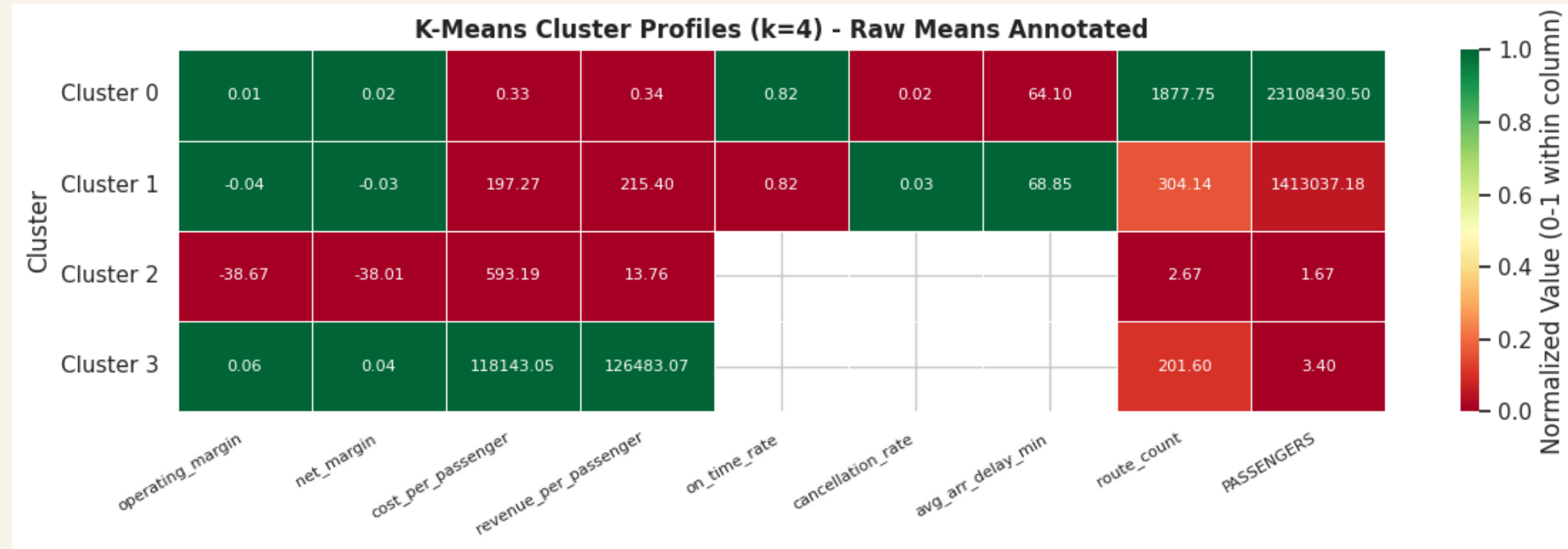
Methodology.

<u>Sub-Question</u>	<u>Outcome</u>	<u>Method</u>	<u>Early Signal</u>
Q1: Cost structure → efficiency	Passengers per route	Comparisons + OLS	Legacy 3–4× higher than LCC/Regional
Q2: Costs → service performance	On-time rate, cancellation rate	Scatterplots + regression	Weak direct relationship
Q3: Business model differences	Carrier-type profiles	Group comparisons + K-means	4 distinct operating clusters
Q4: COVID period shifts	All of the above by period	Period interaction terms	All periods structurally distinct





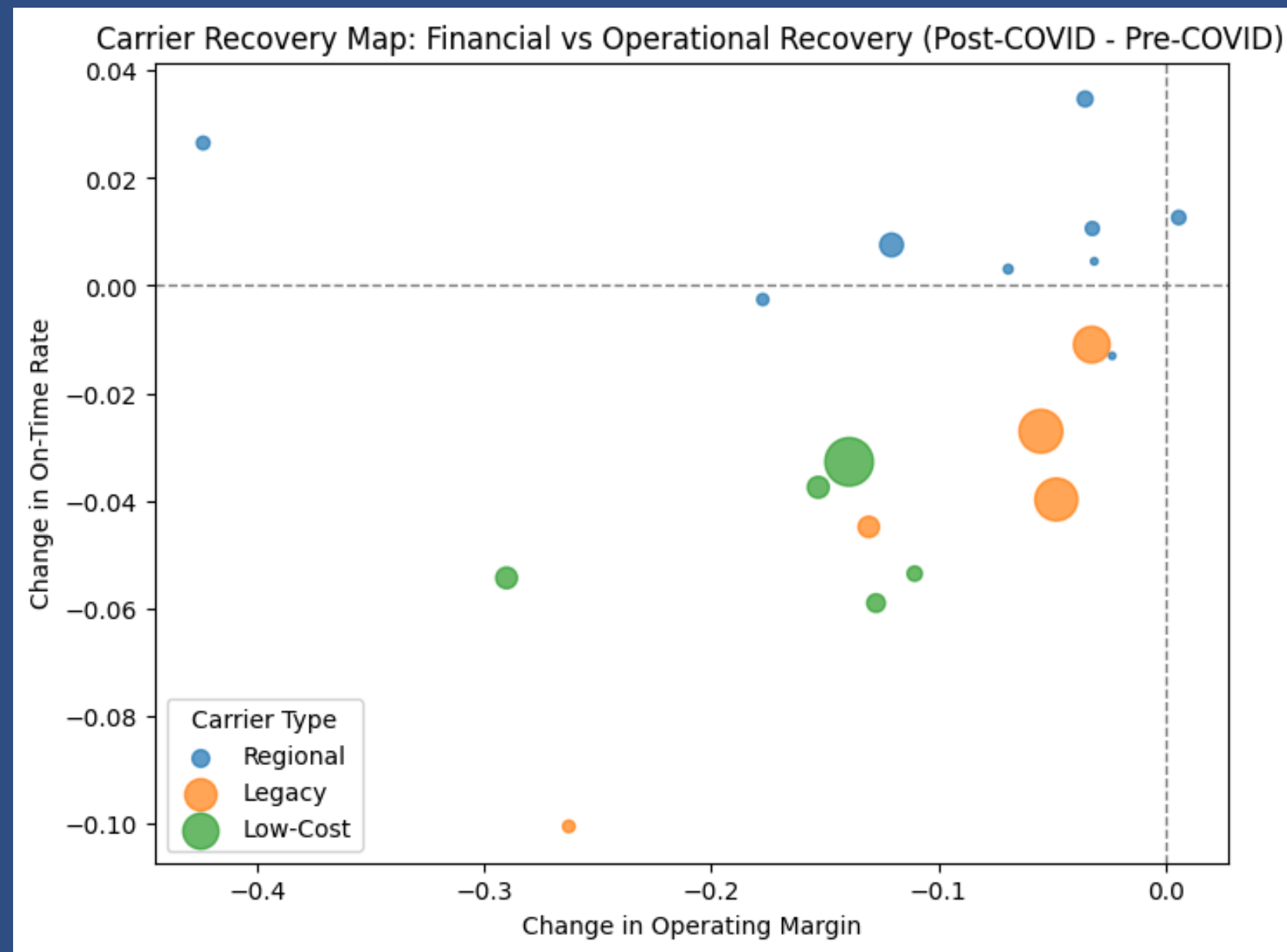
Early Results: Cluster Profiles



- Cluster 0 — Large-scale carriers: 23M+ passengers, 1,877 routes, barely positive margins (1%), on-time rate 0.82. The industry's biggest players.
- Cluster 1 — Mid-scale carriers: 1.4M passengers, slightly negative margins (-4%), similar on-time performance. A large middle tier.
- Cluster 2 — Extreme outliers: margins of -3,867%, fewer than 2 passengers per observation. Likely data anomalies or micro-carriers at the edge of the sample.
- Cluster 3 — Cargo-dominant: cost/revenue per passenger exceeds \$118K-\$126K with ~3 passengers — passenger metrics are meaningless here.
- **Carrier-type labels alone aren't enough — operating profile clusters cut across them**



Early Results: Recovery Map



- Every carrier saw operating margins decline post-COVID vs. pre-COVID — no one fully recovered financially (all x-values < 0)
- But carriers diverged operationally: a few Regional carriers showed slight on-time improvements; most Legacy and Low-Cost carriers declined on both dimensions
- **Financial and operational recovery are not the same thing and do not move together**

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Challenges + Next Steps

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Challenges + Next Steps

Challenges

- Uneven dataset overlap across T-100, Form 41, and OTP (only 681 obs. in all three)
- No seats/departures in T-100 → utilization proxied with passengers per route
- Passenger-based cost ratios are hard to interpret for cargo carriers
- Cluster 2 extreme values need investigation before drawing conclusions

Next Steps

- Run final OLS/regression models with period interaction terms
- Add formal inferential results and significance testing
- Refine carrier classification
- Finalize results and discussion in LaTeX
- Continue grounding findings in the literature





Thank You!

Presented by: Jood Alaswad, Arnol Garcia, Marissa Miller, Brandon Thacker